AI Ethics

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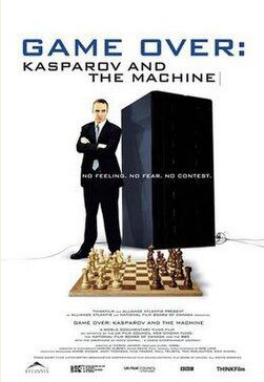
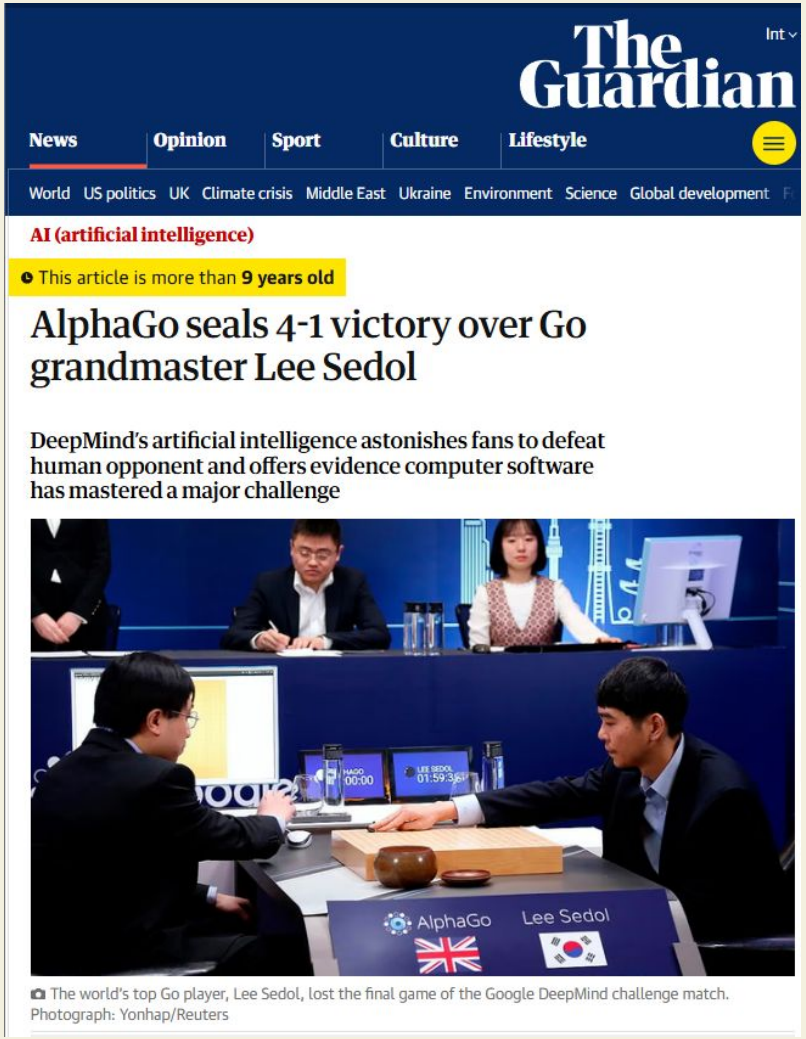


In 1997, chess grandmaster Garry Kasparov resigns after 19 moves in a game against Deep Blue, a supercomputer by IBM. This was the sixth and final game of their match, which Kasparov lost two games to one, with three draws.

From 1984 to 2006, Kasparov was ranked the world's No. 1 player for regular competitive chess.



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AI (artificial intelligence)

This article is more than 9 years old

AlphaGo seals 4-1 victory over Go grandmaster Lee Sedol

DeepMind's artificial intelligence astonishes fans to defeat human opponent and offers evidence computer software has mastered a major challenge

The world's top Go player, Lee Sedol, lost the final game of the Google DeepMind challenge match. Photograph: Yonhap/Reuters

18 times world champion.

"Even if I become the number one, there is an entity that cannot be defeated."

– Lee Sedol , 2019, his retirement from professional play.

Questions to ponder

- If a machine can do things, what is left for us?
- Are we just machines?
- Are we inferior machines, with too many bugs?

(Some) Key Drivers of AI Advancement

- Data Availability
- Competitive Advantage
- National Security and Geopolitics
- Research and Scientific Breakthroughs
- Labor Shortages and Productivity Needs
- Consumer Demand for Innovation
- Economic Potential of the companies

The rush, however, comes with challenges,

- ethical concerns
- the risk of over-dependence on automated systems,

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Faulty update knocks out several Windows computers and servers around the world; IndiGo has cancelled at least 283 trips, and airports around the country saw delays and crowding

Updated - July 20, 2024 06:20 am IST

AROON DEEP

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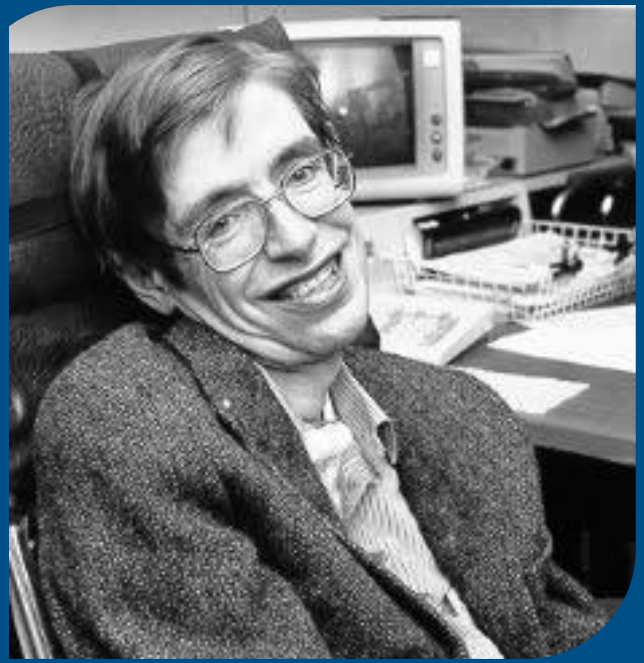
The rush, however, comes with challenges,

- ethical concerns
- the risk of over-dependence on automated systems,
- need for regulation to ensure safe and fair AI development.

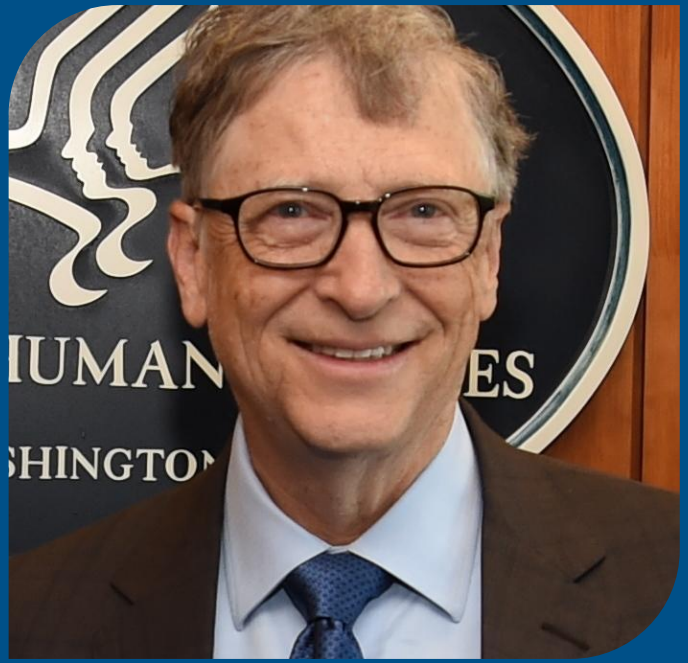
Stephen Hawking

"Success in creating AI would be the biggest event in human history,…"
"Unfortunately, it might also be the last, unless we learn how to avoid the risks. In the near term, world militaries are considering autonomous-weapon
systems that can choose and eliminate targets." "…**humans, limited by slow biological evolution, couldn't compete and would be superseded by A.I.**"

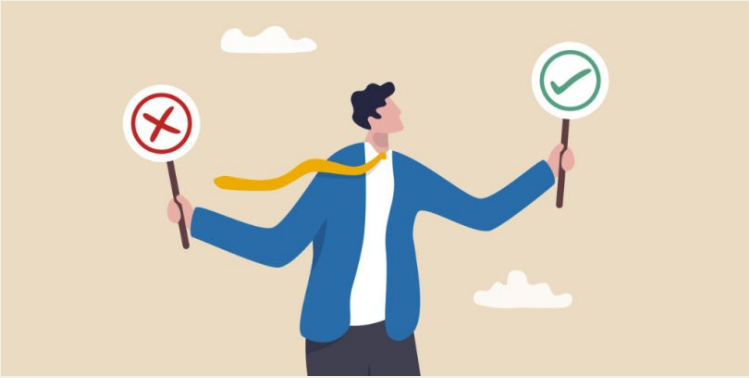


Bill Gates

"I am in the camp that is concerned about super intelligence. First the machines will do a lot of jobs for us and not be super intelligent. That should be positive if we manage it well. A few decades after that though the intelligence is strong enough to be a concern. I agree with Elon Musk and some others on this and **don't understand why some people are not concerned**."



What is Ethics?

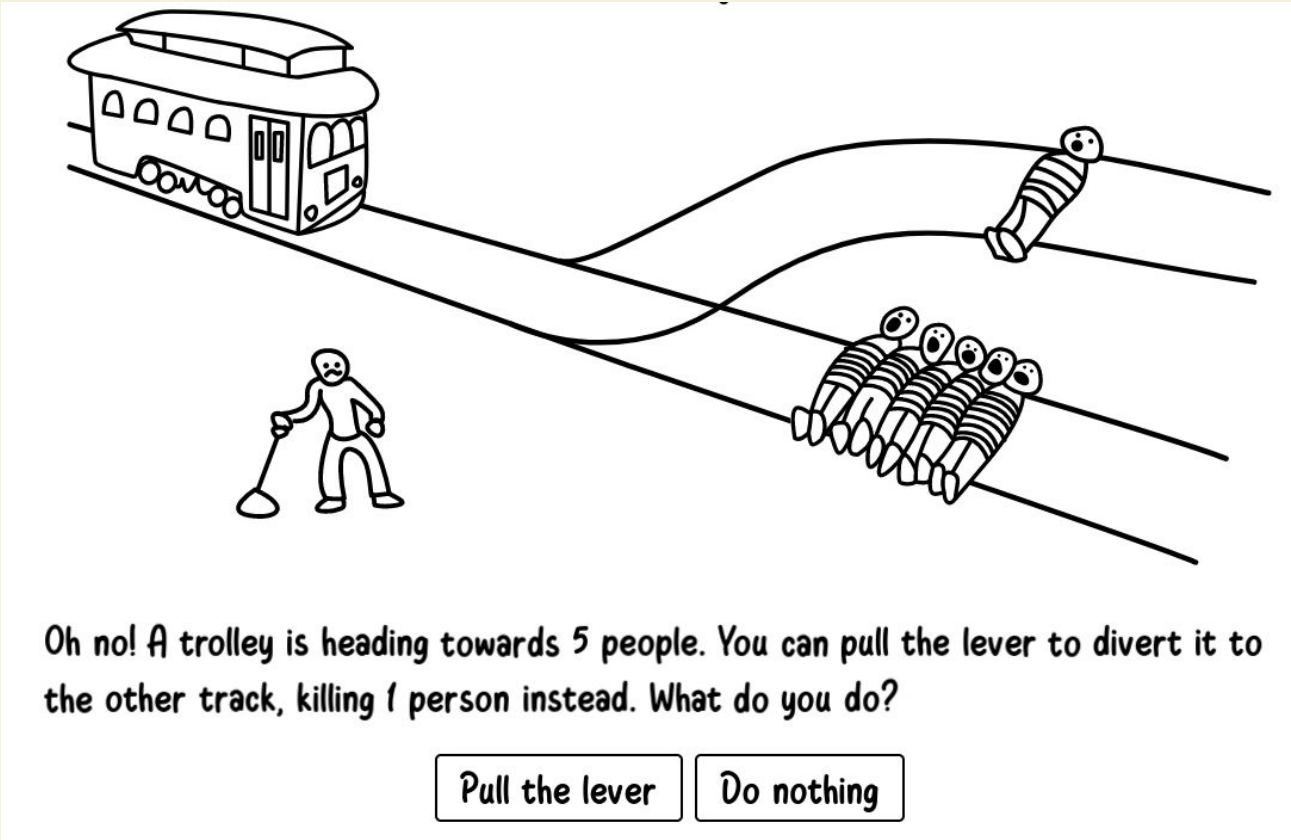


Ethics is a study of what are good and bad ends to pursue in life and what it is right and wrong to do in the conduct of life.

a practical discipline.

how one ought to live and what actions one ought to do in the conduct of one's life.

The Trolley Dilemma



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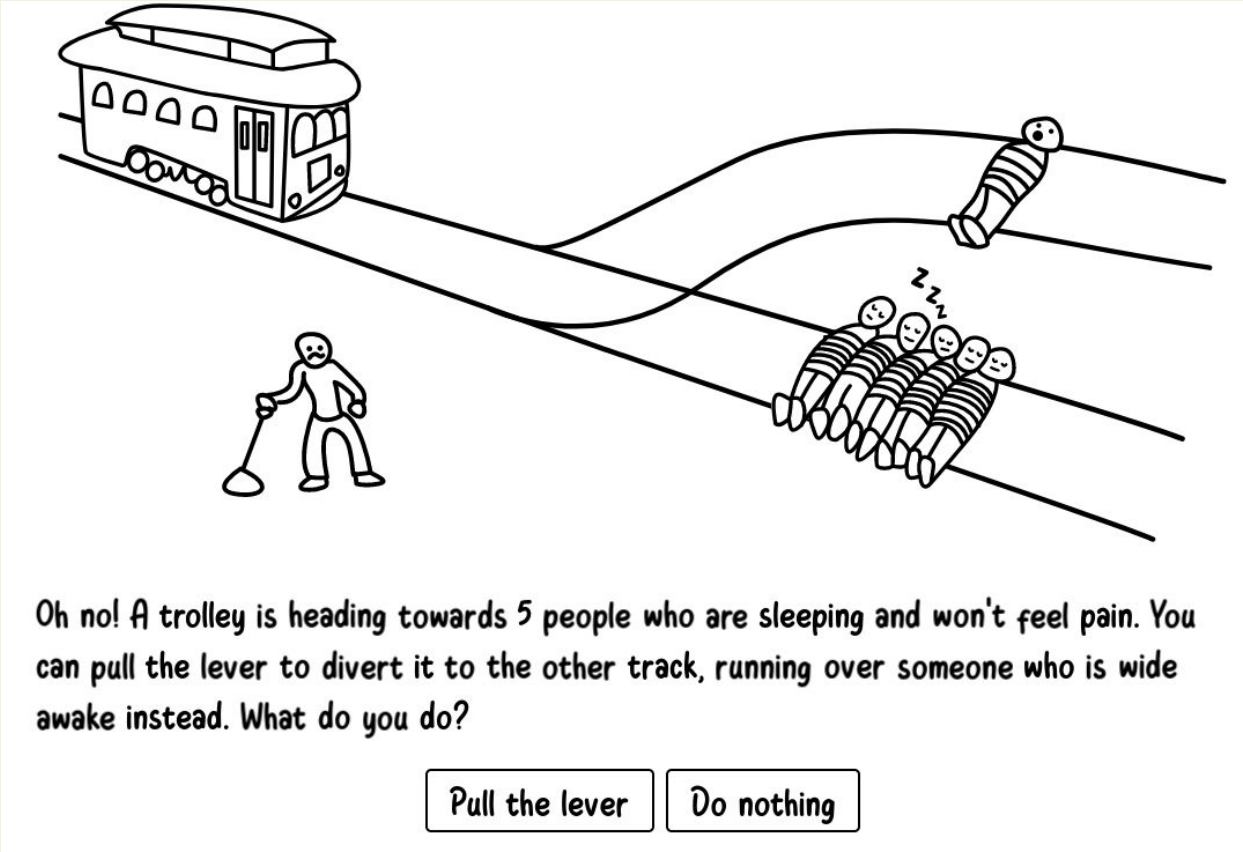

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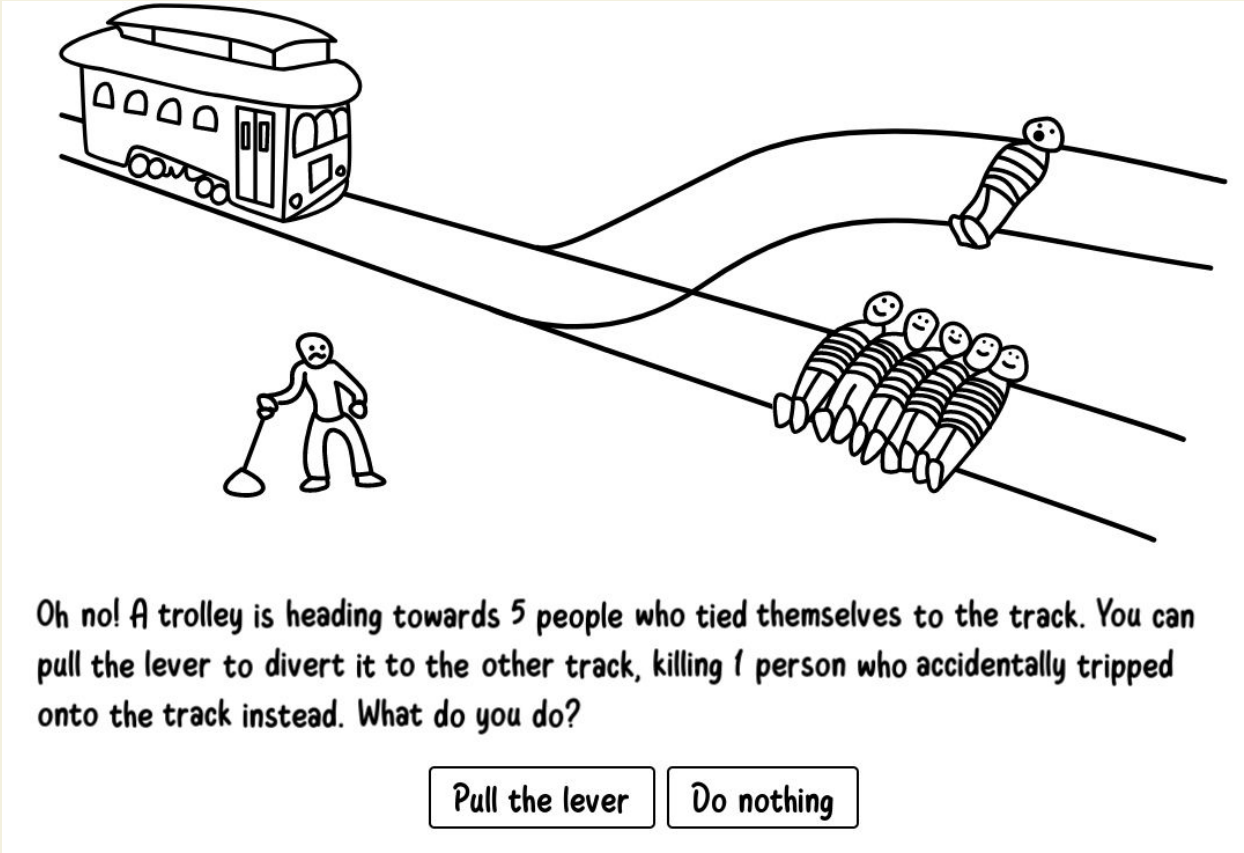

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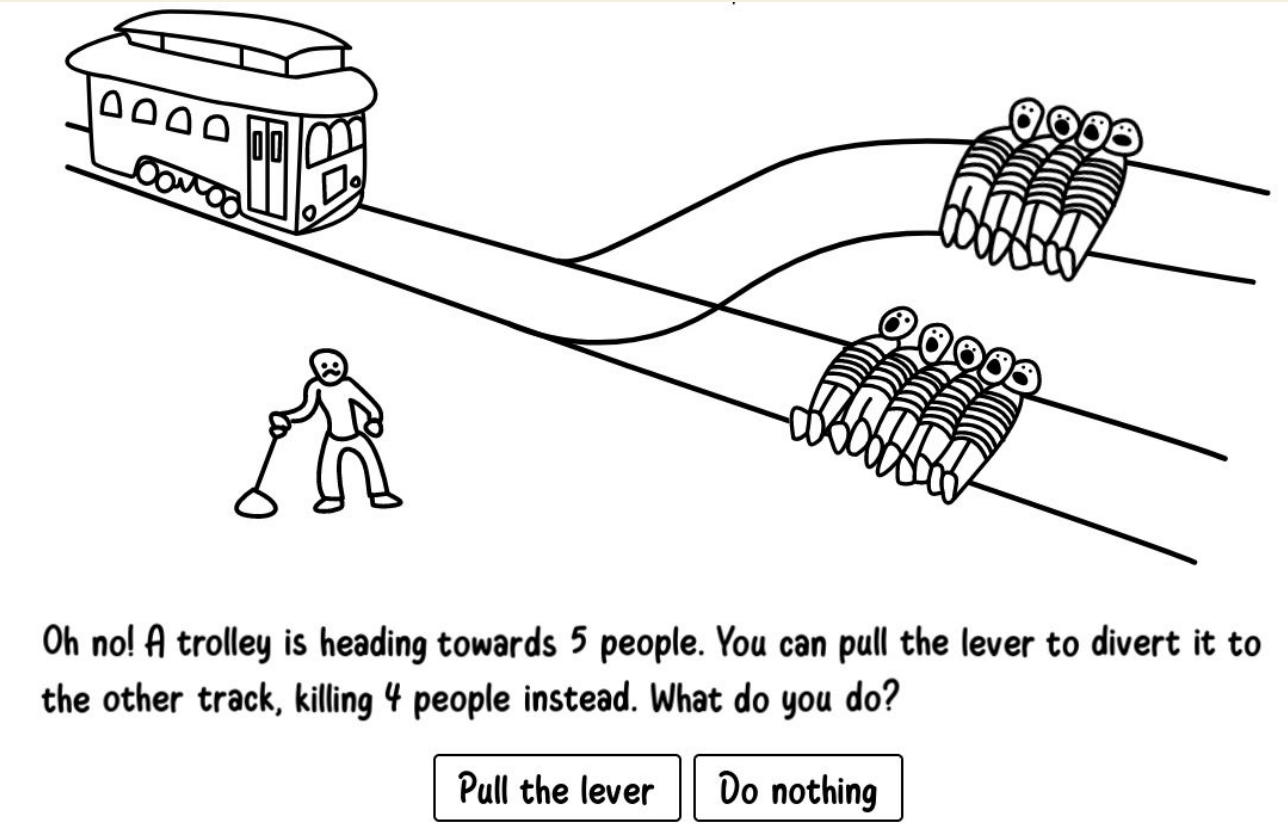

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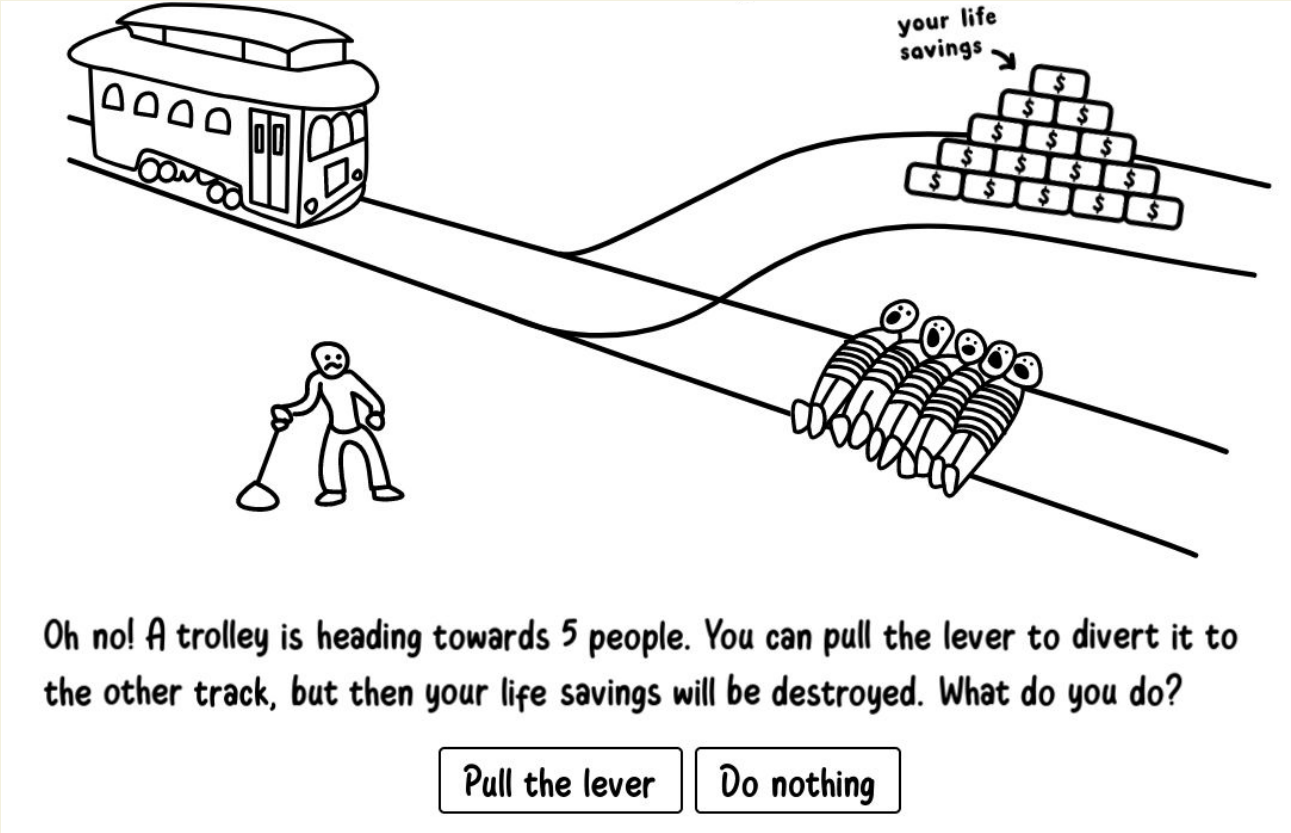

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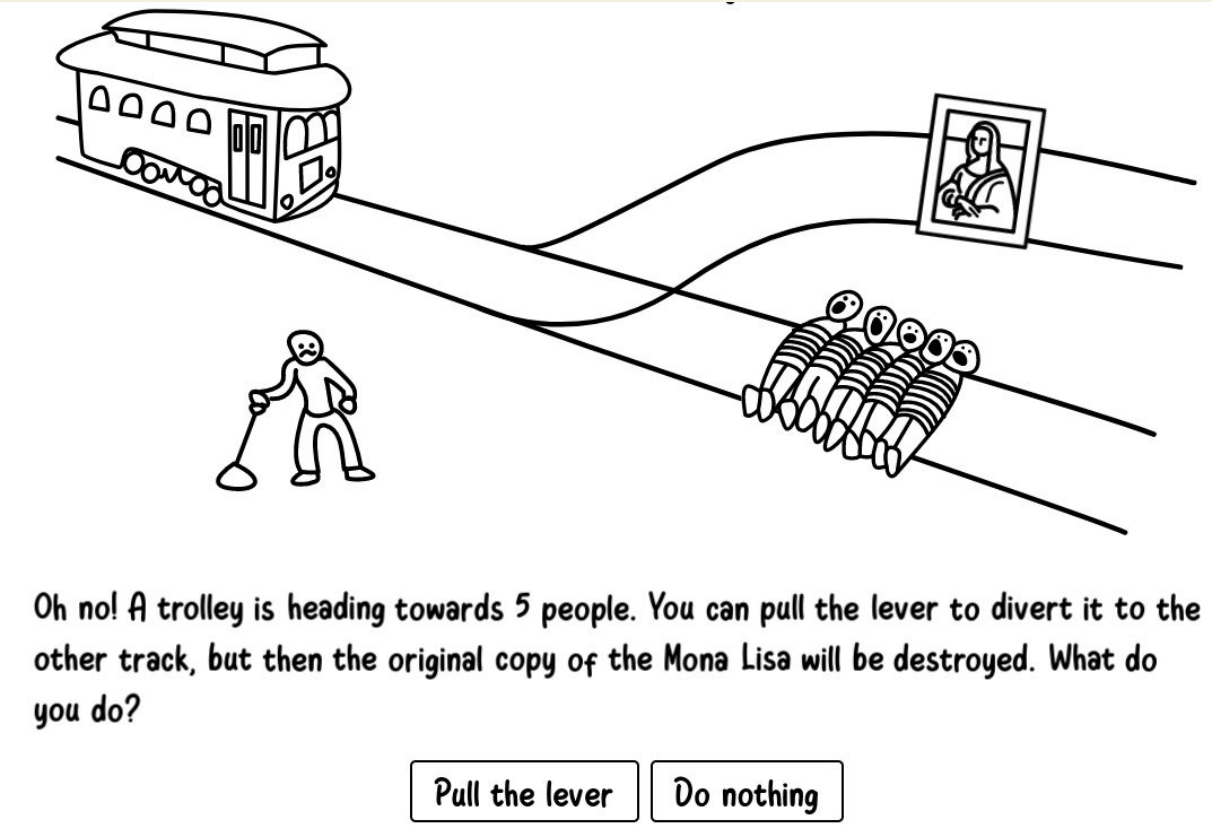

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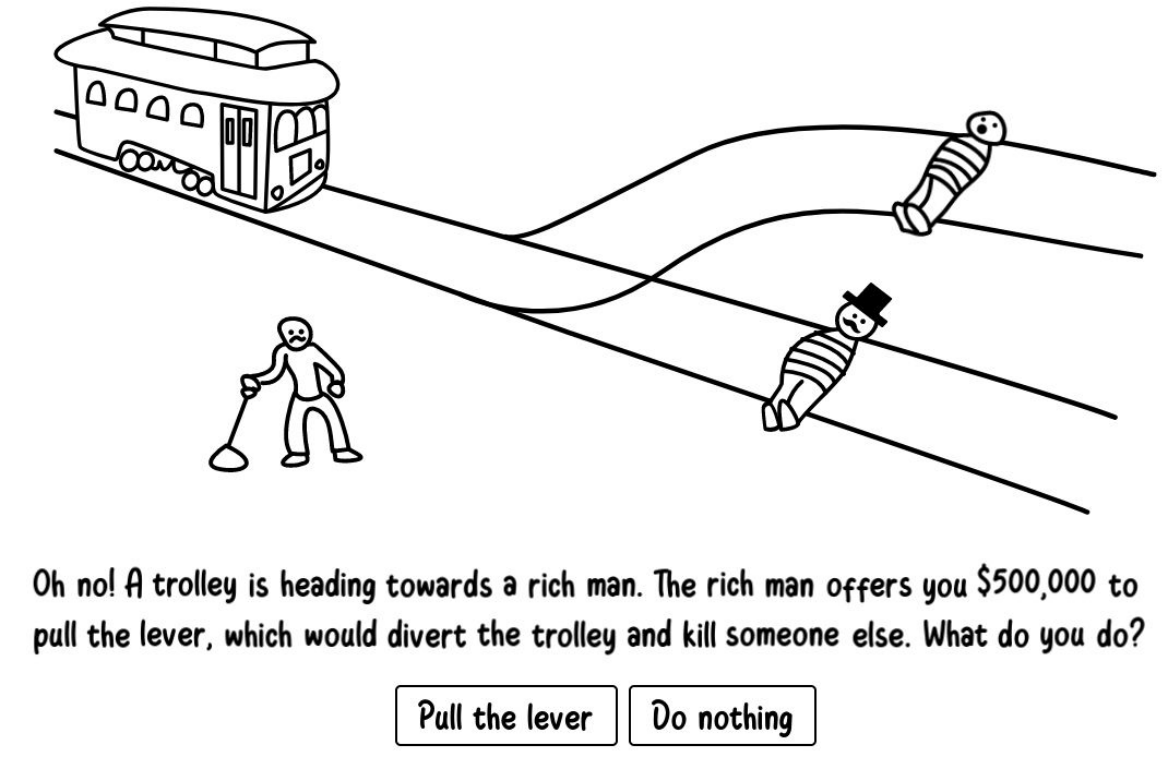

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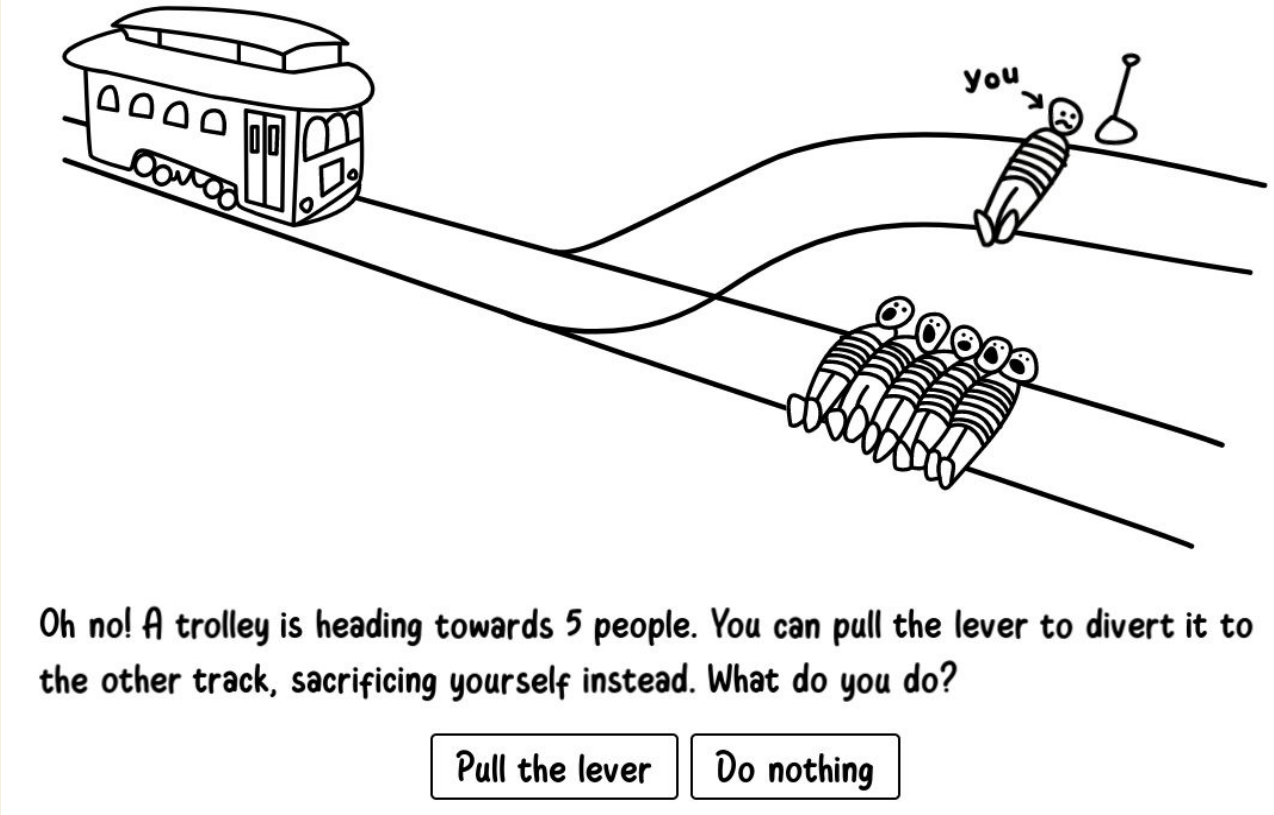

https://neal.fun/absurd-trolley-problems/

What is Ethics?

There are gray areas. **We often don't have easy answers.**

Ethics **changes over time** with values and beliefs of people.

What is AI Ethics?

The Creation of Adam, the first human,

God: "Let us make humankind in our image, according to our likeness".

AI ethics is a set of values, principles, and techniques that employ widely accepted standards of right and wrong to guide moral conduct **in the development and use of AI technologies.**

Questions to ponder

- If a self driving car gets into a situation where it must choose between driving in to a child or into a wall to save the child but potentially killing its passenger, what should it do?

- Should autonomous lethal weapon allowed at all? Who is responsible if something goes wrong?

NEWS

Artificial Intelligence and Machine Learning

AI Researchers Call For Ban on Autonomous Weapons

By Keith Kirkpatrick

Posted Jul 27 2015

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Artificial Intelligence and robotics researchers are asking the United Nations to support a ban on lethal autonomous weapons systems.

More than 1,500 artificial intelligence (AI) and robotics researchers, including visionaries such as Elon Musk and Stephen Hawking, have signed an open letter urging the United Nations (UN) to support a ban on lethal autonomous weapons systems.

The letter follows the April 2015 meeting of the Convention on Conventional Weapons (CCW), which was held at the UN's Palais des Nations in Geneva.

The letter notes that, although in the near future it will be possible to develop and deploy weapons that can be operated

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1 November 2023

First Committee Approves New Resolution on Lethal Autonomous Weapons, as Speaker Warns 'An Algorithm Must Not Be in Full Control of Decisions Involving Killing'

Eight Resolutions on Conventional Weapons, Requiring 21 Separate Votes, Forwarded to General Assembly

Even if an algorithm can determine what is legal under international humanitarian law, it can never determine what is ethical, the First Committee (Disarmament and International Security) heard today after it approved a new draft resolution on lethal autonomous weapons systems (document A/C.1/78/L.56).

An algorithm must not be in full control of decisions that involve killing or harming humans, Egypt's representative said after voting in favour of the resolution. The principle of human responsibility and accountability for any use of lethal force must be preserved, regardless of the type of weapons system involved, he added.

The resolution expresses concern about the possible negative consequences and impact of autonomous weapons systems on global security and regional and international stability, including the risk of an emerging arms race, and lowering the threshold for conflict and proliferation, including to non-State actors.

Content of the course

- Potential Harms caused by AI.
- Why AI can be unethical?
- How to measure bias in AI?
- How to mitigate/avoid those biases?

Lecture sessions, Case studies, Lab tutorials, and guest lectures.

Course logistics

	Mode of assessment	Timeline	Weightage
1	Discussion over ethical concerns on given scenarios in applications of AI (In groups)	Test 1	15%
2	Seminar (In groups)	Test 2	35%
3	Written Exam	End sem	40%
4	Written QnA	At the end of lectures	10%